

Problem A. Little Efresh and Monitor

Input file: standard input
Output file: standard output

Little Efresh found a monitor that shows a single integer x . Next to the monitor, there is a panel with n buttons. Each button applies one of the following operations to the current value of x :

- **and** v_i : after pressing the button, the number on the monitor becomes $x = x \& v_i$
- **or** v_i : after pressing the button, the number on the monitor becomes $x = x | v_i$

Here, $\&$ and $|$ denote bitwise AND and bitwise OR, respectively.

For each query, you know the initial number x shown on the monitor and the required final number y . You may press buttons in any order, and any button may be pressed multiple times.

For each query, determine whether it is possible to obtain y starting from x . If possible, output a sequence of button presses.

* *Bitwise AND* ($a \& b$): the i -th bit of the result is 1 if and only if the i -th bit of a is 1 and the i -th bit of b is 1; otherwise, it is 0.

* *Bitwise OR* ($a | b$): the i -th bit of the result is 1 if and only if the i -th bit of a is 1 or the i -th bit of b is 1; otherwise, it is 0.

Input

The first line contains two integers n ($1 \leq n \leq 2 \cdot 10^5$) and q ($1 \leq q \leq 10^4$) — the number of buttons and the number of queries.

The next n lines describe the buttons. Each line contains a string "**and**" or "**or**" and an integer v_i ($0 \leq v_i \leq 10^5$).

The next q lines contain two integers x and y ($0 \leq x, y \leq 10^5$) — the initial value on the monitor and the value that must be obtained.

Output

For each query output:

–1 if it is impossible to obtain y from x , otherwise

An integer k — the minimum number of button presses, followed by k lines describing the pressed buttons in order.

Each of the next k lines must be exactly one of the following:

- **and** v_k
- **or** v_k

Denoting the pressing of the corresponding button.

Scoring

Subtask	Additional Constraints	Points
0	Examples	0
1	$x_i, y_i, v_j \leq 1$	7
2	$n \leq 100, q \leq 100, x_i, y_i, v_j \leq 1000$	21
3	$n \leq 4000, x_i, y_i, v_j \leq 2000$	16
4	All buttons are of the same type	9
5	$q \leq 10$	28
6	No additional constraints	19

Example

standard input	standard output
3 2 and 3 or 6 and 7 5 3 8 5	3 and 3 or 6 and 3 -1